International Journal of Adolescence and Youth

ISSN: 0267-3843 (Print) 2164-4527 (Online) Journal homepage: www.tandfonline.com/journals/rady20

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To cite this article: Johanna Pfeuffer, Heidrun Stoeger & Albert Ziegler (2025) Amplified social inheritance in STEM: parental careers shape adolescent identity and choice intentions, International Journal of Adolescence and Youth, 30:1, 2551107, DOI: 10.1080/02673843.2025.2551107

To link to this article: https://doi.org/10.1080/02673843.2025.2551107

INTERNATIONAL JOURNAL OF ADOLESCENCE AND YOUTH
2025, VOL. 30, NO. 1, 2551107
https://doi.org/10.1080/02673843.2025.2551107

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Amplified social inheritance in STEM: parental careers shape adolescent identity and choice intentions

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ABSTRACT
Understanding how family environments shape adolescents' STEM pathways is critical for designing effective educational interventions and addressing persistent equity gaps in STEM participation. We test for the existence of an amplified social inheritance effect by distinguishing families with no, one or two STEM parents based on student perception and attribute the (amplified) social inheritance effect to identification with STEM. Survey data were collected from 1,253 secondary school students in Germany (686 identified as female, 567 as male). Participants were predominantly aged 14–15 years. Drawing on the conceptualization of family habitus and STEM identity, family constellations, STEM choice intentions, and students´ identification were analysed using ANOVA, post-hoc comparisons, and measured variable path analysis (MVPA). The results support the existence of an amplified social inheritance effect. STEM identity emerged as a key variable through which parental occupations are associated with students' STEM choice intentions.

ARTICLE HISTORY
Received 25 April 2025
Accepted 18 August 2025

KEYWORDS
Parents; STEM; Identity; choice intention; gender

Introduction

A growing body of research on social inheritance in STEM suggests that children are more likely to pursue science, technology, engineering, and mathematics (STEM) pathways if their parents are employed in these fields: empirical studies have documented systematic differences between families with and without a STEM background regarding educational success and STEM participation (Jacobs et al., 2017; Plasman et al., 2021). These differences highlight the influential role of the family environment in shaping young people's academic identities and career trajectories (Y. Chen et al., 2022; Dou et al., 2019; Šimunović & Babarović, 2021). Building on this foundation, the present study seeks to advance understanding of the mechanisms and conditions underlying the intergenerational transmission of educational and occupational STEM choice intentions. Drawing on student perception on parental occupation, we address two central aims: (1) Does the effect of parental STEM background differ depending on whether one or both parents are

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Supplemental data for this article can be accessed online at https://doi.org/10.1080/02673843.2025.2551107

employed in STEM fields – and how might gender constellations play a role? (2) What psychological mechanism mediates the link between parental occupation and adolescents' outcomes, with a specific focus on STEM identity? At the heart of our inquiry is the examination of an amplified social inheritance effect: adolescents with two STEM-employed parents will demonstrate higher levels of STEM identity and stronger intentions to pursue educational and occupational choices than those with only one or no STEM parent. Building on the concept of family habitus, as introduced by Archer et al. (2012), we propose that family habitus varies systematically by professional background, such that families in STEM occupations foster distinct everyday practices, value systems, and identity pathways that influence children's educational development. For children in these families, such exposure might offer numerous advantages – more frequent informal learning opportunities, stronger competencies, and a more robust identification with STEM domains.

Literature review

The amplified social inheritance effect in light of existing theoretical accounts

Adolescence is a pivotal period for identity exploration, vocational imagination, and the formation of future-oriented goals (Erikson, 1968; Nurmi, 2004). Although early experiences provide the cognitive and motivational foundations for STEM engagement (McClure et al., 2017), it is during adolescence that these experiences are increasingly woven into a developing sense of identity and imagined future pathways. Research shows that STEM identity and career intentions become increasingly differentiated and predictive during this phase (Myint & Robnett, 2024; Zhao et al., 2024). Adolescents are more likely to reflect on their academic strengths, internalize social feedback, and imagine themselves in future professional roles, making this a critical window for interventions that support STEM persistence (Hamilton et al., 2024). Moreover, longitudinal studies suggest that science efficacy and STEM identity mutually reinforce each other during adolescence and jointly predict career aspirations in STEM fields (Zhao et al., 2024). Thus, while early exposure is essential, adolescence is uniquely positioned as a developmental inflection point where identity, motivation, and opportunity converge to shape long-term STEM trajectories. Thus, it is during this phase that parental models, normative expectations, and epistemic framings may have a particularly strong impact on how young people construct viable pathways into STEM. The mechanisms we propose – such as consistent recognition or structural opportunity mapping – are therefore not only theoretically grounded but developmentally timed to the heightened salience of future orientation and social modelling in this life stage. We suggest that during this developmental phase, an amplified social inheritance effect in STEM can be observed: adolescents with two STEM-employed parents exhibit significantly higher levels of both STEM identity and STEM career intentions compared to their peers with one or no STEM-employed parent. This theoretical reasoning is consistent with several theoretical perspectives discussed in the literature.

One relevant mechanism may be the consistency of social recognition adolescents receive in dual-STEM households. STEM identity develops not only from internal interests but also from being seen and treated by others as a 'STEM person'

(Grimalt-Álvaro et al., 2022; Hazari et al., 2010). When both parents are engaged in STEM fields, the likelihood of receiving consistent, reinforcing signals across settings – in conversations, routines, and evaluations – increases. This redundancy can strengthen the internalization of a STEM-oriented self-concept, particularly when mirrored by feedback from peers and teachers.

A second mechanism concerns the density and diversity of informal STEM exposure. Dual-STEM families may offer a broader range of science-related experiences, from dinner-table conversations to shared interests in scientific media, hobbies, or practices. These environments can scaffold familiarity with STEM discourse and problem-solving styles, leading to what Archer et al. (2012) describe as a form of science capital – an accumulation of valued knowledge, practices, and social codes that make STEM domains feel accessible and relevant.

A third mechanism involves normativity. In families where both parents work in STEM, science-related thinking and careers are often portrayed as natural, expected, or default options. This sense of 'what people like us do' reflects the family's habitus (Archer et al., 2012) and can lower the perceived distance between the adolescent's social world and the imagined world of science. In contrast to environments where STEM is framed as exceptional or aspirational, normativity fosters a perception of STEM as personally and socially coherent.

Fourth, epistemic breadth and framing may play a role. When two parents from different STEM subfields contribute to a child's upbringing, they may expose the adolescent to a broader range of ways to think, reason, and problem-solve. Such distributed epistemic framing (Hammer & Elby, 2003) can enable adolescents to recognize multiple legitimate approaches to scientific work, making STEM feel more pluralistic and attainable, even if one route seems misaligned.

A fifth, often-overlooked mechanism is the cultural modelling of cognitive and metacognitive strategies. In STEM-literate households, adolescents may regularly observe and adopt domain-specific ways of asking questions, persisting through uncertainty, or evaluating claims – what some have referred to as cognitive apprenticeships (Brown et al., 1989).

Finally, structural opportunity mapping provides a practical and enabling dimension. STEM-educated parents often have a 'better' insight into the educational system, awareness of high-quality programmes, and social capital to navigate institutional barriers (Lareau, 2003; Xie et al., 2015). They may also possess tacit knowledge about entrance requirements, internships, or scholarship opportunities. This knowledge does not merely open doors – it helps young people imagine those doors in the first place.

Taken together, these mechanisms do not suggest that dual-STEM families deterministically produce children with STEM-oriented interests. Instead, they illustrate how multiple, developmentally aligned factors may converge to make a STEM identity more thinkable, desirable, and actionable during adolescence. Our findings are consistent with this perspective and underscore the need for future research to examine how these factors interact across different cultural and educational contexts. We acknowledge the impact of other social agents like siblings, other relatives and peers, but recognize parents as decisive reference persons in the context of family influence, identity formation and educational trajectories at this stage.

Intergenerational inheritance of STEM occupations

This study builds on empirical findings concerning the intergenerational transmission of STEM choice intentions. Prior research has demonstrated that children of parents employed in STEM fields are more likely to develop STEM-related educational and occupational intentions (Cheng et al., 2019; Chise et al., 2021; Jacobs et al., 2017). These studies have distinguished between different types of STEM occupations, such as maths-intensive versus communication-intensive roles (Cheng et al., 2019), and those with gendered associations—e.g. biology and health (often seen as female-coded) versus engineering, IT and mathematics (male-coded) (Sikora & Pokropek, 2012). Evidence suggests that the intergenerational transmission effect applies to several STEM disciplines. For instance, children are more likely to pursue careers in science, engineering or other STEM subjects like chemistry or statistics if their parents work in these specific fields (Chise et al., 2021). Studies focusing on technical fields, such as engineering, also show a strong inheritance pattern (Anaya et al., 2022; Ardies et al., 2015; Jacobs et al., 2017). However, the magnitude of this effect appears to depend on both the gender of the child and the parent. For example, girls are particularly likely to choose a STEM career if both parents are employed in STEM (Smyk, 2017). Conversely, other studies found that while parental STEM high school major influenced boys more than girls Dahl et al. (2024), others reported the opposite – that parental STEM occupations positively influence girls' intentions more than boys' (Anaya et al., 2022). Slight support exists for the same-sex transmission hypothesis, whereby girls are more influenced by their mothers' STEM careers and boys by their fathers' (Sikora & Pokropek, 2012). In sum, the evidence consistently shows that children from STEM families are more likely to pursue STEM pathways themselves. However, as described above, this pattern is not universal, and the effects vary by gender, parental role, and occupational domain.

Reasons for the comparative advantage of children from STEM families

Several studies have explored mechanisms that may explain why children from STEM families possess a comparative advantage and are addressed in this section. We shortly give an overview on existing differences between families depending on their STEM background that could explain differences in STEM choice intentions and identification with STEM at a later stage. First, children from STEM families achieve better performances in STEM related areas. Research showed, that early numerical competencies are higher among preschoolers whose parents hold STEM qualifications, though not necessarily among those whose parents are actively employed in STEM roles (Mues et al., 2021). Also at a later stage, children from STEM families often perform better in mathematics at school (Shoraka et al., 2016), with gender and grade level moderating these effects (Gutfleisch & Kogan, 2022). This aligns with research on comparative skill advantage, which suggests that children whose parents have strong mathematical skills tend to perform better in maths and are more likely to choose STEM careers (Hanushek et al., 2023). These advantages are not solely the result of subject-specific transmission. Instead, they reflect broader educational benefits associated with having parents in science-related fields (Altmejd, 2023). Second, children from STEM families benefit from different learning environments. For example, children from STEM families benefit from more informal and formal learning opportunities at home (Mues et al., 2021). Mothers in STEM fields, for instance, often demonstrate higher self-efficacy and involvement in children's learning (Zucker et al., 2021). Adolescents with at least one parent in a STEM occupation tend to receive more STEM-related support and exhibit stronger motivational beliefs than their peers without such family backgrounds (Hsieh & Simpkins, 2022). Third, qualitative research indicates that besides fostering familiarity with scientific practices and providing support for their children's career trajectories, parents with STEM careers are role models to their children (Chakraverty & Tai, 2013).

Nevertheless, the influence of parenting appears to be gendered – both on parent and child-level. For instance, Hoferichter and Raufelder (2019) found that girls and boys benefit differently from parental support. While adolescent girls benefit from mother´s support in STEM subjects over time, adolescent boys do not. In contrast, the father´s support was neither for boys´ nor for girls´ academic achievement influential over time. The authors suggest, that girls might perceive parental support differently than boys in the interplay between parental support with parental pressure and that parents behave differently according to the gender of the child, for example in reference to time allocation and type of feedback. Besides, they suggest, that boys might be more influenced by their peers than girls and that boys are interested in STEM, independently from their parents´ support. In deviation from the previous study, Šimunović and Babarović (2021) found that parental support predicts STEM interest, for both boys and girls. Their research supports, that parents might apply different forms of socializing behaviour depending on the child´s gender, providing more STEM-related support to boys than to girls. These results are in line with theoretical reasoning on gendered socialization, since parents might hold different child-specific beliefs and consequently parenting behaviours, depending on the child´s gender, own values and gender-stereotypes (J. Eccles, 2015; J. S. Eccles & Wigfield, 2020).

STEM identity as a key mechanism

While multiple factors contribute to the intergenerational inheritance of STEM, this study posits that a central mechanism is the development of a STEM identity. We argue that children from STEM families are more likely to internalize STEM as part of their self-concept, which in turn drives their educational and career intentions. STEM identity has emerged as a powerful predictor of choice and persistence in STEM domains, even in the face of social barriers (S. Chen et al., 2021; Dou et al., 2019; Godec et al., 2024; Hazari et al., 2010; Seyranian et al., 2018). Family influence, particularly in shaping early science experiences, has been highlighted as a key factor in identity formation. The concept of science capital – science-related attitudes, knowledge, out-of-school learning experiences, and family discourse around science – has proven especially relevant for career aspirations in science, engineering, and physics, in the sense of the desired field of study (Moote et al., 2020). Research by Godec et al. (2024) shows that science capital and science identity predict STEM aspirations, though the potential mediating role of identity in this process remains underexplored. Dou et al. (2019) have explicitly demonstrated that STEM identity mediates the link between early informal experiences (e.g. science talk and media consumption) and STEM career intentions. Moreover, home-based support and science discussions with family members predict key identity indicators, such as perceived competence and recognition in STEM (Dou & Cian, 2022). Longitudinal findings confirm that activities like playing with STEM-related toys or watching science programming in early childhood can have lasting effects on STEM identity, even after accounting for later interest levels. Gender differences in these formative experiences are well documented. Girls are less likely than boys to engage in many of the positive predictors of STEM identity and more likely to experience negative ones (Cohen et al., 2021). Among Hispanic university students, only science talk with family, particularly with parents and siblings, was significantly linked to STEM identity, even when controlling for early science experiences and home support (Dou & Cian, 2021). These findings above suggest that the family environment provides material and cognitive resources and nurtures a deep-seated sense of identity with STEM. This identity, we propose, is a key mechanism through which the intergenerational inheritance of STEM careers is realized.

The current study

This study further examines the social inheritance of occupations, specifically within the STEM field. Building on student data, we assess students´ perception about their parents´ profession and distinguish between families in which neither parent works in the STEM field, families in which the mother works in the STEM field, families in which the father works in the STEM field, and families in which both parents work in the STEM field. The study reflects the subjective assessment of parental occupations rather than an objective categorization and thus differs from studies in which the parents' occupation was surveyed by the parents themselves or was categorized based on exhaustive occupational lists. Using the students´ assessment of parental jobs might provide a different and closer look to students´ realities in their home environment.

As far as we know, two previous studies have directly compared the number of parents working in STEM and its implications for intergenerational transmission. The first focused on the intention to pursue an engineering career and analysed data from U.S. college students (Jacobs et al., 2017), while the second investigated 9th graders' planned occupation by age 30, highlighting different patterns for boys and girls and emphasizing the role of parental stereotypes and children's beliefs (Smyk, 2017). Since Jacobs et al. (2017) demonstrated that the influence of parents varies historically, this study seeks to replicate these findings in a different educational and cultural context and extending prior research by incorporating an identity framework. We assume that advantages stemming from a parental STEM background accumulate with the degree of exposure. This accumulation is expected to manifest in children's identification with STEM and consequently intentions to pursue a STEM pathway. This leads us to the first set of research questions, which focuses on the amplified social inheritance effect:

RQ1a: Do children's STEM choice intentions differ depending on how many parents work in a STEM field?

RQ1b: Do children's identifications with STEM differ depending on how many parents work in a STEM field?

Building on previous research suggesting that the impact of parental background may vary by gender and by which parent works in STEM, we also examine potential interaction effects between family background and gender:

RQ2a: Does the different extent of children´s STEM choice intentions, depending on their parental STEM background, vary by gender?

RQ2b: Does the different extent of children´s identification with STEM, depending on their parental STEM background, vary by gender?

Finally, to explore the mechanism underlying the observed relationships, we ask whether STEM identity mediates the association between parental background and STEM choice intentions:

RQ3: Can the association between the parental STEM background and children's STEM choice intentions be partly explained by higher identification with STEM?

Methodology

Data and sample description

This study is part of a larger German research project aimed at promoting STEM engagement, particularly among girls. The project aims to find out which STEM support programmes are effective in the long term in motivating girls and boys to pursue STEM choices inside, outside and after the school career, taking the different life circumstances of students into account. The project includes a longitudinal online survey starting with either Year 8 or Year 9 students, depending on the duration of secondary education, and continuing until graduation. Usually, students from the age of 14 are addressed in these classes and tracked until they are 18–19 years old. The study involves different types of school (Gymnasium, Gesamtschule and Stadtteilschule). Those school types have in common that their students can take the exam for the university entrance qualification (Abitur) at the end of their school career. All participating schools focused on STEM promotion. This was a criterion for participation in the study. Participation by both schools and students was voluntary. After agreeing to participate, teachers introduced the research project to their classes. Information letters and consent forms for both students and parents were distributed. Teachers actively collected the consent forms. Only students who submitted signed consent forms were permitted to participate. On the day of the survey, each student received an individual access code from the supervising teacher. Surveys were administered via school computers or tablets and took approximately 45 minutes. A supervising teacher was present during survey administration. The study was conducted in German. All questions referred to the German acronym MINT, which stands for the subjects of mathematics, computer science, natural sciences and technology. The term MINT differs from the English term STEM (science, technology, engineering, and mathematics) in that engineering is listed instead of computer science.

The current analysis draws on data from the first wave, collected between October 2022 and July 2023. In total, 51 schools and 1,976 students from 184 classes participated in this initial measurement point. 239 observations were excluded from the analysis due to missing proof of declarations of consent, implausible information, and response patterns. Cases with missing data on key variables were removed. Most missing cases can be attributed to the recoding of parental STEM background, since 245 students could not classify their mother´s and 324 students could not classify their father´s occupation as a STEM or non-STEM job. Though we would have wished to analyse 26 people with diverse genders as a further subgroup, we needed to exclude them due to small subgroup size, as well as 11 students from German schools abroad for analytic clarity. Further information on missing data is provided in the supplementary material. The final sample consisted of 1,253 students which are spread across 166 classes from 48 schools. Of these, 567 identified as male ($M_{age} = 14.40$, $SD = 0.89$), and 686 as female ($M_{age} = 14.22$, $SD = 0.86$). 415 students had at least one parent that was born abroad and 178 students reported that German was not the most frequently spoken language in their household.

Measures

Parental STEM background

Students were asked whether they would describe their mother's and father's occupations as a typical STEM profession (e.g. maths teacher, biologist, engineer). The questions read:

> Would you describe your mother's job as a typical STEM profession?
>
> Would you describe your father's job as a typical STEM profession?

Each question included a note to accommodate diverse family constellations: 'If you do not know what applies to your mother, please state what applies to your first legal guardian,' and correspondingly for the father. Response options were 'yes,' 'no,' and 'don't know.' Students who answered 'don't know' were treated as missing. Responses were recoded into a composite variable with four categories:

> Neither parent works in a STEM field
>
> Only the mother works in a STEM field
>
> Only the father works in a STEM field
>
> Both parents work in a STEM field

Although the students dispose preknowledge about the STEM field, they may not have a shared understanding of STEM professions. It is possible that certain occupations were categorized both as STEM and as non-STEM professions depending on the students' perception of STEM work. The operationalization thus represents the perception of students as to whether their parents' profession can be classified as a STEM profession, rather than a clear-cut operationalization based on a priori defined categories. This procedure is similar to the approach of Ardies et al. (2015) who captured the technological background of parents assessing whether the job of the father or the mother 'has something to do with technology' (Ardies et al., 2015, p. 51).

Assuming, that students´ answers take into account prior knowledge, impressions, and experiences, this operationalization opens up a different perspective on parental STEM occupation and students´ reality, that is the subjective perspective of adolescents.

In the case of same-sex parents, the gender of the parents was not recorded correctly, what might also apply partly to the information provided about legal guardians. These misclassifications pose a threat to the validity of the measurement of the parents' gender. Due to the low number of families with same-sex parents and children growing up outside their families (Federal Ministry for Family Affairs, Senior Citizens, Women and Youth, 2024; Statistisches Bundesamt, 2023), we assume that distortions exist but are very low. Further information on family compositions in Germany are provided in the limitations.

STEM choice intentions

STEM choice intention was measured using six items adapted from the Academic Elective Intentions Scale by Stoeger et al. (2013). The scale included items measuring both educational choice intentions, like in-school, out-school and curricular choices, and occupational intentions. Each item began with the prompt: *'I can picture myself . . . '* Example items are:

> I can picture myself majoring in a STEM subject.
>
> I can picture myself taking part in activities like for example STEM competitions or olympics.
>
> I can picture myself taking a job that has to do with STEM.

All six items of the scale are presented in the supplementary material. Students rated their agreement on a six-point Likert scale (1 = not true at all, 6 = completely true). A mean index score was calculated, with higher scores indicating stronger STEM choice intentions. Cronbach's alpha was .93 for the overall sample and both gender subgroups.

Identification with STEM

STEM identification was assessed using an adaptation of the Science Identity Scale developed by Vincent-Ruz and Schunn (2018). Students indicated their agreement with the following four items:

> I am a STEM person.
>
> My family thinks of me as a STEM person.
>
> My friends think of me as a STEM person.
>
> My teachers think of me as a STEM person.

In line with the broader project, the original four-point scale was extended to a six-point scale (1 = not true at all, 6 = completely true). A composite index was created, with higher scores indicating stronger STEM identification. Internal consistency was high ($\alpha = .90$ overall; $\alpha = .89$ for males; $\alpha = .91$ for females).

Gender

Gender was assessed via the question: 'What is your gender?' Response options included male, female, and diverse. For analytical purposes, only responses from students identifying as male or female were included.

Data analysis

All analyses were done using the statistical software Stata, version 18 (StataCorp, 2023).

Analysis of variance

To answer the first and second set of research questions, we conducted a series of analysis of variance (ANOVA) tests. Before running these models, we tested the variance homogeneity and normality assumptions. Levene's test was used to assess homogeneity of variance. For the outcome variable *Identification with STEM*, Levene's test indicated variance heterogeneity between female and male students in the group with two STEM parents. Following Blanca et al. (2018), we calculated the variance ratio (largest variance/smallest variance), which was below the threshold of 1.5, allowing us to proceed with the F-test.

Normality of residuals was assessed using the Shapiro-Wilk test. While normality was violated in a few subgroups, ANOVA is robust to such violations when group sizes are unequal (Blanca et al., 2017; Schmider et al., 2010). Therefore, we conducted:

- One-way ANOVAs to test differences in STEM choice intention and identification across the four categories of parental STEM background
- Post-hoc comparisons using Tukey's Honestly Significant Difference (HSD) test
- Two-way ANOVAs to test for interaction effects between gender and parental STEM background

Path analysis

A mediation model was calculated to analyse the relationship between the parental STEM background and the STEM choice intentions (see Aichholzer, 2017, p. 51). Structural equation modelling is based on the assumption that an a priori postulated model and the empirical data fit. The parameter estimation, therefore, aims to achieve the best possible fit of the observed covariance matrix S and the model-implied covariance matrix $\Sigma(\theta)$ (Aichholzer, 2017, p. 107f.). The maximum likelihood method is used for parameter estimation. As the variables were defined as manifest variables, only structural models and no measurement models were estimated. The theoretical model on which the path analyses are based is shown graphically in Figure 1. We decided against the inclusion of covariates in the model, but depict the path model with the covariates SES, migration background and gender in the supplementary material. The maximum likelihood method provides for categorical dependent variables and a multivariate normal distribution. For this reason, the standard errors are estimated using the Satorra-Bentler method. The method estimates corrected standard errors and the Satorra – Bentler scaled chi-squared test for the structural equation model (Aichholzer, 2017, p. 113f.). To estimate the structural model, its identification is required. The model meets the t-rule $t \leq p(p + 1)/2$, where t corresponds to the number of model parameters to be estimated and p corresponds to the number of

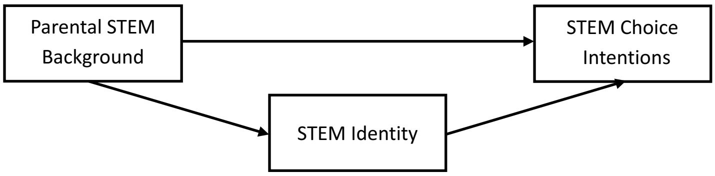

Figure 1. The hypothesized mediation model with STEM identity as the mediator of the relationship between parental STEM background and students' STEM choice intentions.

independent and dependent variables (Urban & Mayerl, 2014, p. 76). In order to assess multicollinearity, the variance inflation factor was calculated separately for each regression equation. For both regression equations, the average VIF was about 1.1, with 1.09 being the highest value. The VIF does not indicate multicollinearity.

Results

Descriptive statistics

With regard to the main variables of analysis, the descriptive statistics show that 58.82% of the respondents reported that neither parent works in a STEM profession. In 6.54% of the cases, students reported that the mother works in STEM; in 26.82% of cases, students reported a STEM occupation of the father; and in 7.82% of cases, students indicated that both parents were employed in STEM. The average score on the STEM identity scale was 3.23 (SD = 1.37), while the average score for STEM choice intention was 3.10 (SD = 1.21). Table 1 provides an overview of descriptive statistics for all analysis variables, disaggregated by gender. The correlation between the study variables as well as descriptive statistics on migration background and parental SES are shown in the supplementary material.

Research question 1a

Research Question 1a examined whether students' STEM choice intentions differ depending on their parental STEM background. A one-way analysis of variance was conducted to

Table 1. Descriptive statistics.

Variables	Categories	Whole sample				Female subgroup				Male subgroup			
		n	%	M	SD	n	%	M_f	SD_f	n	%	M_m	SD_m
Parental STEM Background	Both parents do not work in the STEM field	737	58.82%			391	57.00%			346	61.02%		
	Mother works in the STEM field	82	6.54%			45	6.56%			37	6.53%		
	Father works in the STEM field	336	26.82%			194	28.28%			142	25.04%		
	Both parents work in the STEM field	98	7.82			56	8.16%			42	7.41%		
STEM Identity				3.23	1.37			3.00	1.35			3.51	1.36
STEM Choice Intention				3.10	1.21			3.02	1.22			3.20	1.20

Note. N = 1253 (n_f = 686, n_m = 567).

Table 2. ANOVA results of STEM choice intentions by parental STEM background.

	Sum of Squares	*Df*	Mean Square	*F*	Prob $> F$
Between groups	87.84	3	29.28	20.80***	< .001
Within groups	1757.89	1,249	1.41		
Total	1845.73	1252	1.47		

*$p < .05$; **$p < .01$; ***$p < .001$.

Table 3. Pairwise comparisons of means with equal variances for STEM choice intentions.

Pairwise comparisons			Contrast	SE	$P > \|t\|$	95% CI
Mother works in the STEM field	vs.	Both parents do not work in the STEM field	0.36*	0.14	.047	[0.00, 0.71]
Father works in the STEM field	vs.	Both parents do not work in the STEM field	0.37***	0.08	< .001	[0.17, 0.57]
Both parents work in the STEM field	vs.	Both parents do not work in the STEM field	0.89***	0.13	< .001	[0.56, 1.22]
Father works in the STEM field	vs.	Mother works in the STEM field	0.01	0.15	1.00	[−0.36, 0.39]
Both parents work in the STEM field	vs.	Mother works in the STEM field	0.53*	0.18	.016	[0.07, 0.99]
Both parents work in the STEM field	vs.	Father works in the STEM field	0.51**	0.14	.001	[0.16, 0.86]

*$p < .05$; **$p < .01$; ***$p < .001$.

address this question. The results are presented in Table 2. The model yielded an explained variance of 29.29 and an unexplained variance of 1.41. The F-ratio indicated statistically significant differences in STEM choice intentions between the groups. The effect size was $\eta^2 = .05$, corresponding to a medium effect.

Post-hoc comparisons (see Table 3) revealed significant differences between students who indicated to have no STEM parent and those who reported either a STEM mother, a STEM father, or two STEM parents. Additionally, students who indicated two STEM parents reported significantly higher choice intentions than those who indicated to have a STEM mother or father. No significant difference was observed between the groups that indicated to have a STEM mother and a STEM father, suggesting similarly elevated levels of STEM choice intention in both groups.

Research question 1b

Research Question 1b explored differences in STEM identity depending on the parental STEM background. Again, a one-way ANOVA was used (see Table 4). The model accounted for an explained variance of 38.07 and an unexplained variance of 1.80. The F-ratio showed significant group differences in STEM identity scores. The effect size was $\eta^2 = .05$, indicating a medium effect.

Table 4. ANOVA results of STEM identity by parental STEM background.

	Sum of Squares	Df	Mean Square	*F*	Prob $> F$
Between groups	114.21	3	38.07	21.13***	< .001
Within groups	2250.11	1,249	1.80		
Total	2346.33	1252	1.89		

*$p < .05$; **$p < .01$; ***$p < .001$.

Table 5. Pairwise comparisons of means with equal variances for STEM identity.

Pairwise comparisons			Contrast	SE	$P > \|t\|$	95% CI
Mother works in the STEM field	vs.	Both parents do not work in the STEM field	0.56**	0.16	.002	[0.15, 0.96]
Father works in the STEM field	vs.	Both parents do not work in the STEM field	0.46***	0.09	< .001	[0.23, 0.68]
Both parents work in the STEM field	vs.	Both parents do not work in the STEM field	0.94***	0.14	< .001	[0.57, 1.31]
Father works in the STEM field	vs.	Mother works in the STEM field	−0.10	0.17	0.93	[−0.53 0.32]
Both parents work in the STEM field	vs.	Mother works in the STEM field	0.38	.20	.23	[−0.13, 0.90]
Both parents work in the STEM field	vs.	Father works in the STEM field	0.49**	.15	.009	[0.09, 0.88]

*$p < .05$; **$p < .01$; ***$p < .001$.

As shown in Table 5, post-hoc tests revealed significant differences in identity scores between students from STEM and non-STEM households. No significant difference emerged between the groups that reported a STEM mother and a STEM father, nor between those with a STEM mother and those with two STEM parents. However, the difference between the groups that reported a STEM father and two STEM parents was significant. This analysis result might suggest that perceiving a STEM mother could be similarly influential as perceiving two STEM parents in shaping STEM identity. However, this non-significant difference should be interpreted with caution given the relatively small sample sizes in these groups, and does not imply statistical equivalence.

Research question 2a

Research Question 2a examined whether the influence of parental STEM background on STEM choice intentions differs by student gender. A two-way ANOVA was conducted, first without and then with an interaction term (Tables 6 and 7, respectively).

While the model without the interaction term showed significant group differences in STEM choice intentions by gender and parental STEM background, the inclusion of the interaction term changed the significance of the main effects. Although the interaction term was not significant, the effect of gender on STEM choice intentions became insignificant after the interaction term was added to the model. The effect of parental STEM background on choice intentions remained insignificant. Although the interaction term itself was not statistically significant, the shift in the gender effect underscores the complexity of the relationships. The overall effect size for the interaction model was

Table 6. Two-way ANOVA results of STEM choice intentions by parental STEM background and gender without interaction term.

	Partial Sum of Squares	*Df*	Mean Squares	*F*	Prob > *F*
Model	100.13	4	25.03	17.90***	< .001
Gender	12.28	1	12.28	8.78**	.003
Parental STEM Background	90.09	3	30.03	21.47***	< .001
Residual	1745.60	1248	1.40		
Total	1845.73	1252	1.47		

* $p < .05$; ** $p < .01$; *** $p < .001$.

Table 7. Two-way ANOVA results of STEM choice intentions by parental STEM background and gender with interaction term.

	Partial Sum of Squares	*Df*	Mean Squares	*F*	Prob > *F*
Model	107.86	7	15.41	11.04***	< .001
Gender	1.11	1	1.11	0.80	.372
Parental STEM Background	86.30	3	28.77	20.61***	< .001
Gender × Parental STEM Background	7.73	3	2.58	1.85	.137
Residual	1737.87	1245	1.40		
Total	1845.73	1252	1.47		

* $p < .05$; ** $p < .01$; *** $p < .001$.

$\eta^2 = .06$, with most of the variance explained by parental STEM background. Based on these findings, we conclude that the group differences in STEM choice intention by parental STEM background do not differ significantly by gender.

Research question 2b

Research Question 2b focused on whether the relationship between parental STEM background and students' identification with STEM varies by gender. A two-way ANOVA with interaction term was conducted (see Table 8). Both main effects were significant, whereas the interaction term approached but did not reach statistical significance ($p = .053$). Thus, the assumption of differing group effects by gender is not supported. The overall model explained a moderate proportion of variance, with an effect size of $\eta^2 = .09$.

Research question 3

Research Question 3 addressed whether STEM identity statistically mediates the relationship between parental STEM background and students' STEM choice intentions. A path analysis was conducted to test this mediation model (see Figure 2). Three dummy variables were created to represent the categories of parental STEM background: mother in STEM, father in STEM, and both parents in STEM. The reference category was students with no STEM parent.

In the regression predicting STEM choice intention, only the group with two STEM parents showed a significant direct association ($\beta = 0.27$) when controlling for STEM identity. Neither perceiving a STEM mother nor a STEM father alone had a significant direct association with STEM choice intentions once identity was included. In contrast, in the regression predicting STEM identity, all three dummy variables had significant positive associations with STEM identity: STEM mother ($\beta = 0.56$), STEM father ($\beta = 0.46$), and two

Table 8. Two-way ANOVA results of STEM identity by parental STEM background and gender with interaction term.

	Partial Sum of Squares	*Df*	Mean Squares	*F*	Prob > *F*
Model	214.03	7	30.58	17.70***	< .001
Gender	23.75	1	23.75	13.75***	< .001
Parental STEM Background	115.80	3	38.60	22.35***	< .001
Gender × Parental STEM Background	13.33	3	4.44	2.57	.05
Residual	2150.30	1245	1.73		
Total	2364.33	1252	1.89		

*$p < .05$; **$p < .01$; ***$p < .001$.*

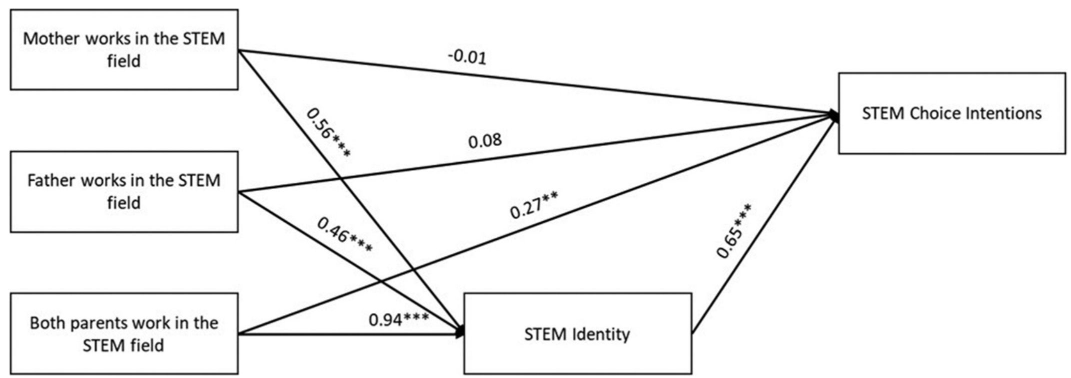

Figure 2. Visualization of the path model; *p < .05; **p < .01; ***p < .001.

STEM parents (β = 0.94), with the latter being the strongest. For students who reported a STEM mother the entire association between parental STEM background and choice intentions could be attributed to STEM identity. For children who reported a STEM father a large part of the association (β = 0.30) could be attributed to STEM identity. For students who reported two STEM parents, an association with STEM choice intentions of 0.28 remained, after controlling for identity. Table 9 summarizes the results of the mediation model.

In order to account for potential bias due to parental SES, the children´s gender and migration background, we performed additional analysis with these variables as covariates. However, the inclusion of these covariates did not meaningfully alter the results of the path model. We provide the extended models in the supplementary materials for complete transparency.

Summary and discussion

This study contributes to research on social inheritance in the STEM field. Based on cross-sectional data of secondary school students in Germany, we showed that children with a family STEM background, which was measured based on the subjective assessment of the parents' occupation as a STEM profession, are on average more likely to intend a specialization in STEM, than students who do not have the impression that their parents work in the STEM field. With regard to specialization, we referred to specialization during the school career or as a career or study choice after graduation. The novel contribution of our study is, that we looked on the one hand at differences of exposure and distinguished families without STEM background, one STEM parent and two STEM parents and on the other hand emphasized associated differences in the subject-related identity of students with STEM. In the course of the study, we posed and answered three research questions.

First, regarding existing group differences in STEM choice intentions and STEM identity, we observed a cumulative effect: children that reported one STEM parent demonstrate stronger STEM identification and intention to choose STEM than children that reported no STEM parent. Furthermore, children that reported two STEM parents show on average higher identification values and STEM choice intentions than children that reported one STEM parent. This pattern supports the concept of an amplified social inheritance effect. Our findings align with those of Jacobs et al. (2017), who demonstrated students are more likely to pursue a career in engineering when both parents are employed in the STEM field, compared to having only one parent in such a profession, but diverge from Smyk (2017), who identified gender differences in the influence of parental STEM occupations. An attempt to explain the differences between the studies could be made by including the parents' working hours, time spent at home, time spent with the children and children´s time spent in informal childcare or education system on either country or individual level. For example OECD Data shows differences in the earning patterns of couples with children in the United States and Germany (OECD, 2025). It is also possible that differences in findings reflect variations in sample size and measurement between studies rather than substantive differences in effects.

Second, when considering the child's gender, no significant interaction effects were found between gender and parental STEM background concerning STEM choice intentions or identification. This suggests that both boys and girls equally benefit from a

Table 9. Results of the mediation model.

Model	Parameter	Coefficient	Satorra-Bentler SE	$P>\|z\|$	95% CI
Direct Effects on STEM Choice Intentions					
	STEM Identity	0.65***	0.02	< .001	[0.62, 0.69]
	Mother works in the STEM field	−0.01	0.09	.949	[−0.18, 0.17]
	Father works in the STEM field	0.08	0.05	.151	[−0.03, 0.18]
	Both parents work in the STEM field	0.27***	0.09	.003	[0.09, 0.45]
Direct Effects on STEM Identity					
	Mother works in the STEM field	0.56***	0.15	< .001	[0.27, 0.85]
	Father works in the STEM field	0.46***	0.09	< .001	[0.28, 0.63]
	Both parents work in the STEM field	0.94***	0.14	< .001	[0.66, 1,22]
Indirect Effects on STEM Choice Intentions					
	STEM Identity				
	Mother works in the STEM field	0.36***	0.10	< .001	[0.17, 0.56]
	Father works in the STEM field	0.30***	0.06	< .001	[0.18, 0.41]
	Both parents work in the STEM field	0.61***	0.09	< .001	[0.43, 0.80]
Indirect Effects on STEM Identity					
	Mother works in the STEM field				
	Father works in the STEM field				
	Both parents work in the STEM field				
Total Effects on STEM Choice Intentions					
	STEM Identity	0.65***	0.02	< .001	[0.62, 0.69]
	Mother works in the STEM field	0.36**	0.13	.006	[0.10, 0.61]
	Father works in the STEM field	0.37***	0.08	< .001	[0.22, 0.53]
	Both parents work in the STEM field	0.89***	0.13	< .001	[0.64, 1.14]
Total Effects on STEM Identity					
	Mother works in the STEM field	0.56***	0.15	< .001	[0.27, 0.85]
	Father works in the STEM field	0.46***	0.09	< .001	[0.28, 0.63]
	Both parents work in the STEM field	0.94***	0.14	< .001	[0.66, 1.22]

$^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$.

parental STEM background, contrasting with studies that have reported gender-specific parental influences (Anaya et al., 2022; Cheng et al., 2019). The existence of country differences regarding gender-specific parental influence was already evident in the study by Sikora and Pokropek (2012) who found heterogenous patterns for countries that participated in the PISA study in 2006. For Germany, the authors found gender effects, that diverge from our results: differentiating between more feminine dominated STEM disciplines, like BAH (biology, agriculture, health) and more masculine dominated STEM disciplines, like CEM (computing, engineering and mathematics), the authors found, that boys´ BAH-choice intentions were associated with (mother´s and) father´s BAH occupation, depending on covariates in the model, while for girls only the mother´s BAH occupation was associated with choice intentions. In contrast, for CEM-subjects, only the father´s occupation mattered for boys and girls but not the mother´s CEM occupation (Sikora & Pokropek, 2012). Since we do not distinguish between CEM and BAH, we can speculate whether the combination of the two strands yields our results or whether circumstances have changed over time. In any case, the distinction between STEM fields might provide further insights in the explanation of gendered parental effects.

Third, regarding the mediation, STEM identity could explain a large part of the variance between parental STEM background and students' choice intentions for all three groups, indicating that children´s identity advantages might be of major importance for the social inheritance in the STEM field.

Based on theoretical reasoning on the family habitus, that encompasses shared dispositions, resources, practices, and values that shape what an individual considers 'thinkable, possible, and desirable' (Holmegaard et al., 2024, p. 5), we suggested that families differ systematically with regard to parents´ occupational background. For us, the occupation of parents in the STEM field was worth studying, because it reflects a prominent area of interest within a family, the existence and availability of topic-related knowledge and experience, easier access to a field through formal knowledge or informal opportunities, access to the code of conduct and showing a certain type of career as a natural and feasible option. In particular, we coined the term amplified inheritance effect, to describe the cumulative effect of STEM exposure in the home environment which refers to the fact that STEM topics, ways of working and thinking and leisure activities become more present and normative when both parents have a STEM background. Overall, the amplified inheritance effect suggests a multiplicative impact of informal STEM learning experiences within the family through daily conversations, modelling of scientific thinking, and the cultivation of epistemic curiosity. In the data, we found confirmation for our theoretical considerations that the parents´ professional activity, respectively the perception thereof, is associated with the psychological concept of STEM identity. This speaks for systematic differences in socialization by the parents depending on their professional activity. For example, adolescents that reported two STEM parents showed STEM identity scores nearly one standard deviation higher than those who did not report STEM parents. This substantial difference suggests that dual-STEM family environments provide a strong foundation for developing a STEM self-concept, comparable to or exceeding the effects of many targeted STEM enrichment programs. It has also been shown that there is a gap between families with one or two STEM parents. This speaks in favour of the amplified inheritance effect, that theoretically assumed, STEM-related topics, behaviour, values and thinking would be more dominant with two STEM parents. Although not taking explicit family interactions into account, we found that children differ significantly, indicating to intrafamilial processes that shape persons that are more likely to see oneself as a STEM person and to get recognized as STEM person by others, possibly on the basis of children's ´ expressed interest, activities or achievements (Carlone & Johnson, 2007; Hazari et al., 2010). The explicit processes that take place within families was not the emphasis of this study and is not reflected in our dataset. Instead, we aimed to draw attention to the unequal access to the self-image of being a STEM person, based on the recognition of oneself and of others as a STEM person. Since we, as other studies, found strong associations between being a STEM person and STEM choice intentions (Dou et al., 2019; Hazari et al., 2010; Starr et al., 2020), we suggest STEM identity to be a pivotal point for aspiring STEM pathways, that is not equally accessible. Failure to address the different chance on STEM identification risks creating what Pfeuffer et al. (2025) call a 'Sneaky Equity Gap': unintended inequalities that grow unnoticed and can significantly undermine life chances. In the sense of detaching STEM choices from factors of origin, we suggest to consider STEM identity as a modifiable factor. Fostering STEM identity among students, particularly those without familial ties to STEM fields, educators and policymakers can work towards creating more equitable pathways into STEM careers. Therefore, STEM education programs are faced with the task and challenge of appealing to a diverse group of young people, creating physical access to STEM, and making it imaginable, even for those not *born with STEM*. That the parental home is associated with participation in out-of-school education is visible in a recent study by DeWitt et al. (2025) who found that students with STEM parents and with high cultural capital are more likely to participate in informal science learning. To address the selectivity of participants, they suggest to work with communities or organizations to a dress a broader target group. Besides, educators can build on existing studies to foster recognition, for example regarding teaching strategies that make students feel recognized in the classroom (Hazari & Cass, 2018). The amplified inheritance showed that advantages within a family can cumulate with exposure. This provides insight into deep-seated inequality dynamics, but contains still an outlook and positive notion for educators. The differences between families with STEM backgrounds show that the *amplification* makes a difference both in identification as well as in STEM choice intentions. Parents are not the only actors that might provide implicit or explicit STEM content and support in daily life.

Limitations and future research

Notable limitations of this article refer to the measurement of the independent variable. Although at the beginning of the questionnaire, the students were familiarized with the STEM definition and supervising teachers were available during the survey, who could help the students if anything was unclear to them, the students´ classification of parental occupations as STEM-related depended on their own perceptions and prior knowledge. To avoid further effort for teachers and students, we refrained from introducing or providing an additional list of STEM professions before data collection and only provided examples of STEM professions, including engineering and the teaching profession, Thus, different interpretations of what is a STEM occupation could have occurred. Therefore, we clearly mention, that we measured student perception of their parent´s STEM careers. This measurement of parental STEM background differs from the operationalization of other studies in whose research context we locate our study (f.e. Jacobs et al., 2017; Smyk, 2017) and thus limits comparability. According to German administrative data, about 26% of employees that are subjects to social insurance contributions work in the STEM field, including engineering (Bundesagentur für Arbeit, 2025). In contrast to administrative data, we counted teachers with a STEM background as holding a STEM occupation (Bundesagentur für Arbeit, 2022) and found that parents with a STEM background are overrepresented in our sample. On the one hand, this could signalize to inaccurate classification of STEM occupations in reference to objective categorizations. On the other hand, the overrepresentation could be due to our special sample that focused on schools with a focus on STEM promotion. In the sense of social inheritance of STEM, it is plausible that parents with STEM occupations are more likely to choose secondary schools for their children with a focus on STEM. Besides, we need to acknowledge, that a big number of students was not able to classify the occupations of both parents. Further analysis revealed that not-knowing how to classify a STEM profession was linked to a family's migration background and SES children. Besides, children with separated parents may have struggled to classify the occupation of both parents, leading to them being disproportionately often considered missing. However, we cannot estimate the impact of single parenting on our analysis. To address this issue, the supplementary material contains analyses including the students who only provided one answer and adding SES, migration background and gender as additional covariates in the model. Since the changed sample and model did not alter the conclusions of the analysis, we suggest that the core results of our analyses are robust. Given the theoretical entanglement of SES and parental STEM occupation, treating SES and migration background as a mere confounder would obscure rather than clarify the mechanisms under investigation. For the sake of clarity and conceptual coherence, we present the model without covariates in the main text.

With regard to the second research question, the analysis of gender constellations, we need to point to potential ambiguity in determining the gender of parents, as students were instructed to report on the STEM occupations of their legal guardians if unaware of their biological parents' professions. Incorrect assumptions about the gender of the legal guardians were included in the analysis for students with same-sex parents as well as possibly for students whose legal guardians are not the parents. To be transparent about the expected extent of incorrectly coded gender, we would like to provide some insight into official statistics on families in Germany. Based on administrative data, 14.3 million minor children lived in Germany in 2022. 99.4% of them lived in families. In these statistics, the term family covers a broad range of family constellations but is based on parent(s) and children living in the household. Regarding families with two married parents with children under 18, that represents 69% of the families in Germany, in about 0.4% of the families, the parents shared the same gender. The share was higher in families with two unmarried parents and children under 18. This family constellation represents12% of the families in Germany, of those, 1.1% were same-sex couples (Federal Ministry for Family Affairs, Senior Citizens, Women and Youth, 2024; Statistisches Bundesamt, 2025). Although we anticipate that a large part of the students could report data on their parents and their parents´ gender, we must remark that not all children's living environments in terms of the socializers´ genders were recorded correctly. Although we assume that distortions are small, due to the number of cases reported in German administrative data, this drawback of our study equally raises questions about social inheritance with mixed-sex parents or what STEM access exists for children who do not live with their parents. As the additional information on legal guardians is only a note in the questionnaire to allow children from diverse family structures to classify their home environment, it is not possible to estimate the proportion of people who have provided information on legal guardians.

Although we could point to identification advantages based on family background which may be related to the intergenerational inheritance of STEM pathways, our analysis is based on cross-sectional-data, that limits the ability to draw causal conclusions and to assess the development of STEM identity and choice intentions over time. Longitudinal studies are necessary to understand how and when children develop or alter their STEM choice intentions, particularly in relation to parental influence, since parental impact diminishes during adolescence as peer influence becomes more significant (Abels & König, 2016). Additionally, we acknowledge that parental STEM occupation and SES are theoretically entangled. While our supplementary analyses including SES as a covariate showed that the amplified inheritance effect remains robust, this entanglement limits causal interpretation. Our findings may also reflect broader social class advantages associated with higher parental education and occupational status, and generalizability is therefore restricted to similar SES distributions. To account for the nestedness of the dataset, we calculated intraclass correlation coefficients (ICCs) based on a three-level random intercept model, finding very low school-level clustering (0.7% for STEM choice intention; near zero for STEM identity) and modest class-level clustering (8.7% for STEM choice intention; 5.6% for STEM identity). Given these small ICCs, the average class size below 10, and concerns about reduced statistical power with multilevel models in this context, we opted for single-level models to maintain analytic clarity and sample size. We acknowledge that this decision may underestimate standard errors slightly, but believe the substantive conclusions remain robust.

Our study did not account for the gender-typicality of specific STEM professions. Previous research indicates that certain STEM fields are perceived as more masculine or feminine, influencing students' career choices (Cheng et al., 2019; Sikora & Pokropek, 2012). Future research should explore how the gendered nature of parental STEM occupations affects children's identification with STEM and intentions to pursue STEM. Differences between families can be measured by far more factors than professional activity. Lastly, we did not examine other familial factors influencing STEM participation, such as parental stereotypes, values, or beliefs about their children's abilities. The Situated Expectancy-Value Theory posits that these factors significantly impact children's motivation and achievement (J. S. Eccles & Wigfield, 2020). Incorporating these variables in future studies could provide a more comprehensive understanding of the intergenerational transmission of STEM choice intentions.

Acknowledgments

This publication resulted from the joint project „FösaMINT – Förderung schulisch-außerschulischer MINT-Kooperation mit Genderschwerpunkt". The following institutions are practice partners in the project: CyberMentor, Deutsche Telekom-Stiftung, Körber-Stiftung, matrix GmbH, MINT-EC [Nationales Excellence-Netzwerk für Schulen], MINTvernetzt. The responsibility for the content of this publication lies with the authors.

I acknowledge the use of Grammerly (v1.2.154.1654) and DeepL for providing language improvement suggestions. None of the content in this document was generated by AI.

Disclosure statement

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Funding

The project is funded by the Federal Ministry of Education and Research (BMBF) under the project grant number 16MF1091.

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Author contribution

JP conceived of the study, participated in its design, performed the statistical analysis and interpretation of the data, and drafted the manuscript, AZ conceived of the study, participated in its design and funding acquisition, assisted in the interpretation of the data and drafted the manuscript, HS participated in the study design and funding acquisition, helped to draft the paper, and made revisions and critiqued the output for important intellectual content. All authors have read and approved the final version of the manuscript.

Ethics approval

The local legislation and institutional requirements did not require ethical review and approval by an ethics committee for the study on human participants. Our concept for protecting participants' data was based on national standards. It was approved by the participating institutions (e.g. the ministries of participating German states and the principals of participating schools).

Consent to participate

The participants and their legal guardians provided written informed consent to participate in this study.

Consent to publish

As part of the informed consent, participants were informed that the data would be used for scientific purposes and that only anonymized data would be published.

Data availability statement

Datasets are available on request: The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

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